

Deterministic geometric Savitzky-Golay dynamics estimator

Shared parameter update: reference -> estimated

mass [kg]	2.351597591 -> 2.364632518	delta=+0.0130349
ratio=1.005543		
CoG x [m]	-0.002024708562 -> -0.02352122798	delta=-0.0214965
ratio=-		
CoG y [m]	-3.052657894e-05 -> -0.02789348744	delta=-0.027863
ratio=-		
CoG z [m]	0.009509749599 -> 0.04764610998	delta=+0.0381364
ratio=-		
rotor effectiveness 1	1 -> 1.035146159	delta=+0.0351462 ratio=1.0351462
rotor effectiveness 2	1 -> 0.9648821916	delta=-0.0351178 ratio=0.96488219
rotor effectiveness 3	1 -> 0.8896399833	delta=-0.11036 ratio=0.88963998
rotor effectiveness 4	1 -> 0.9512055204	delta=-0.0487945 ratio=0.95120552
rotor command lag [s]	0.004997730255 -> 0.1991368504	delta=+0.194139
ratio=-		

Inertia matrix [kg m^2]

reference:

```
[ 0.06500006148, -7.27899253e-07, 1.901508003e-05]
[-7.27899253e-07, 0.06495265634, 5.9167305e-08]
[ 1.901508003e-05, 5.9167305e-08, 0.1289921107]
```

estimated:

```
[ 0.06512708618, -0.0008175500882, -0.0001420797489]
[-0.0008175500882, 0.0659404496, -0.0006452595023]
[-0.0001420797489, -0.0006452595023, 0.1287785655]
```

Principal moments: reference -> estimated

principal moment 1 [kg m^2]	0.06495264516 -> 0.06461734077	delta=-0.000335304
ratio=0.99483771		
principal moment 2 [kg m^2]	0.06500006701 -> 0.06644328904	delta=+0.00144322
ratio=1.0222034		
principal moment 3 [kg m^2]	0.1289921163 -> 0.1287854715	delta=-0.000206645
ratio=0.998398		

Optimization

joint dynamics loss: 846.862243704
Gaussian-prior cost: 14.3924950007
selected strict-ZOH rotor lag [s]: 0.199136850355

Bag failure_1_single

reconstruction position RMSE [m]: 87.1362207479
reconstruction position component RMSE [m]: [19.354507 9.685932 84.405609]
reconstruction orientation RMSE [deg]: 150.393038615
gyro consistency RMSE xyz [rad/s]: [44.135527 41.947155 23.628531]
specific-force consistency RMSE xyz [m/s^2]: [50.710708 97.068457 44.225745]
inferred external force RMS xyz [N]: [9.847148 10.089174 9.335729]
inferred external torque RMS xyz [N m]: [4.634756 6.591425 9.523612]
bag dynamics loss: 846.862243704
gimbal command lag [s] 0.004998683929 -> 0.01939889572

Recorded command timestamp intervals (data-derived)

Bag failure_1_single

rotor: median dt=0.00499773026 s (200.091 Hz), min=2.00271606e-05 s, max=0.0138893127 s

gimbal: median dt=0.00499868393 s (200.053 Hz), min=4.529953e-06 s, max=2.48282003 s

Selected optimizer diagnostics

status=0 success=False nfev=80 njev=6
message=The maximum number of function evaluations is exceeded.
reported optimality=1.2624563623830909e-05
initial cost=861.254740620267
final cost=861.2547387045346
cost reduction=1.915732354973443e-06
coordinate step L2=0.0008234398350711468
final gradient inf-norm=1.2624563623830909e-05
gradient tolerance satisfied=False
final Jacobian rank=14
final Jacobian cond(nonzero)=430.4615570220279
max directional finite-difference Jacobian relative error=4.636641855461156e-12